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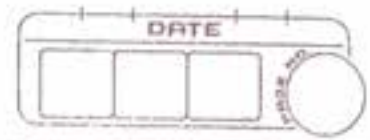
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Subject: operating system.

School / College : _____

[illegible]



Assignment No. 1.

9. Write the assignment on the types of applications. Each operating systems types should contain the points (Definition, Usage, Advantages, Disadvantages).

1. Batch operating system.

Defn: A Batch operating system executes a group (batch) of similar jobs together without requiring user interaction during execution.

usage:- Payroll processing.
Bank transaction processing.
Billing systems.
large data processing.

Advantages:- Efficient for repetitive tasks.
can process large volume of data.
Reduces idle CPU time.
Requires minimal user intervention.

Disadvantages:- No direct interaction with users.
Long waiting time for job completion.
Difficult to detect & fix errors.

2. Time-sharing operating.

Defn:-

A Time-sharing operating system allows multiple users to access the computer simultaneously by sharing CPU time among them.

usage :- multi-user computer system

University computer labs

online servers.

office networks.

Advantages :- supports multiple users at the same time
fast response time.

Efficient CPU utilization.

Easy resource sharing.

Disadvantages :- Requires powerful hardware
security risks due to multiple users

system failure affects all users.

complex to manage.

3. ~~M~~ulti programming operating system :-

Defⁿ :-

A Multiprogramming operating system keeps multiple programs in memory & executes them one after another to maximize CPU utilization.

usage :- Business application

Database servers

Scientific computing.

Advantages :- Better CPU utilization

increase system efficiency.

Reduces CPU idle time.

Disadvantages :- complex memory management,
Difficult scheduling
Requires more memory.

4. Multitasking operating system.

Defⁿ :- A multitasking operating system allows a user to run multiple applications simultaneously.

Usage :- personal computers
Smartphones
Laptops.

Advantages :- Run multiple applications at once
Save time
Improves productivity
Better user experience.

Disadvantages :-
Uses more RAM & CPU
Performance may slow with many tasks.
Risk of software conflicts.

5. Multiprocessing operating system.

Defⁿ :- A Multiprocessing operating system uses two or more processors to execute multiple processes simultaneously.

Usage :- scientific research.
Engineering applications
Data centers.

DATE

Advantages : Increased reliability
Faster processing
Better performance
Handles large workloads efficiently

Disadvantages : Expensive hardware
Complex system design
Higher maintenance cost

6. Real-Time operating system (RTOS)

Defⁿ :- A Real-Time operating system processes data & responds within a fixed time limit.

Usage :- Medical equipment
Air traffic control
Robotic.

Advantages : Very fast response
High reliability
Predictable performance

Disadvantages : Expensive to develop
limited multitasking
Complex programming

7. Distributed operating system.

Defⁿ :- A Distributed operating system manages multiple computers connect through a network & makes them appear as a single system.

usage :- cloud computing
Distributed databases.
Research institutions
large organizations.

Advantages :- Resource sharing.
High reliability
faster processing.

Disadvantages :- complex implementation.
Network dependency.
Security challenges.

8. Network operating system (NOS)

Defⁿ :-

A Network operating system manages network resources & allows computers to communicate & share files, printers & other resources.

usage :- office network
school & colleges
Banks

Advantages :- centralized management.
Easy file & printer sharing.
Better security.

Disadvantages :- costly setup.
 server failure affects clients
 Requires skilled administration.

9. Embedded operating system

Defⁿ :- An Embedded operating system is designed for dedicated devices that perform specific tasks.

Usage :- Smart TVs.
 washing machines
 ATMs
 IoT devices.

Advantages :- Small & lightweight.
 fast performance
 Low power consumption

Disadvantages :- limited functionality
 Difficult to upgrade
 limited memory.

10. Mobile operating system.

Defⁿ :- A mobile O.S. is designed specifically for smartphones, tablets & other portable devices.

Usage : Smartphones

Tablet

Smart watches

Mobile application.

Advantages :- User friendly interface.

Supports Touch Screens

• optimized battery usage

Large app ecosystem.

Disadvantages :- Limited hardware upgrades

security risks from malicious apps.

Battery dependency

limited multitasking compared to desktop o.s.

* Types of System calls

Definition

System calls are the interface between a user program and the operating system kernel. They allow programs to request services from the operating system.

1) Process control system calls.

These system calls are used to create, execute, terminate & manage processes.

2) File Management system calls.

These system calls are used to create, open, read, write, rename & delete files.

3) Device Management system calls.

These system calls allow a program to communicate with hardware devices such as printers, disk, keyboard & displays.

4) Memory Management system calls.

These system calls allocate, free & manage memory for process.

5) Information Maintenance system calls.

These system calls retrieve or modify system information such as process ID, priorities, time & file attributes.

c) Communication system calls.

These system calls enable communication between processes through pipes, shared memory, sockets, signals & message queues.

d) Protection & Security system calls.

These system calls control access, permissions, user authentication & system security.

* Process Management system calls.

1) fork()

Creates a new child process by duplicating the calling process.

2) vfork()

Creates a child process that shares the parent's address space until it calls exec() or exits.

3) clone()

Creates a new process or thread with customizable resource sharing.

4) execve()

Replace the current ^{process} image with a new program.

5) exit()

Terminates the calling process after performing cleanup operations.

6) `_exit()`

Immediately terminates the process without performing standard library cleanup.

7) `wait()`

Suspends the parent process until one of its child process terminates.

8) `waitpid()`

Waits for a specific child process to change its state or terminate.

9) `waitid()`

Waits for child process state changes & provides detailed status information.

10) `kill()`

Sends a signal to process or a group of processes.

11) `getpid()`

Returns the process ID (PID) of the calling process.

12) `getppid()`

Returns the parent process ID (PPID) of the calling process.

13) `nice()`

Changes the scheduling priority of the calling processes.

14) `set priority()`

sets the Scheduling priority of a process, process group or user.

15) `get priority()`

Returns the Scheduling priority of a process, process group or user.

16) `open()`

opens an existing file or creates it & returns file descriptor

17) `create()`

creates a new file or truncates an existing file & opens it for writing.

18) `close()`

closes an open file descriptor

19) `read()`

Read data from a file or device into a buffer

20) `write()`

writes data from a buffer to a file or device

21) `pread()`

Reads data from a specified file offset without changing the file pointer.

- 22) `fwrite()`
Writes data to a specific file offset without changing the file pointer.
- 23) `fseek()`
Moves the file pointer to a specific position in a file.
- 24) `stat()`
Returns information about file using its file path name.
- 25) `fstat()`
Returns information about a file using its file descriptor.
- 26) `lstat()`
Returns information about a symbolic link instead of the target file.
- 27) `access()`
Checks whether the calling process has permission to access a file.
- 28) `unlink()`
Deletes a file by removing its directory entry.
- 29) `rename()`
Changes the name or location of a file or directory.

- 30) `mkdir()`
Creates a new directory.
- 31) `rmdir()`
Removes an empty directory.
- 32) `dup()`
Creates a duplicate copy of an existing file descriptor.
- 33) `dup2()`
Duplicates a file descriptor to a specified descriptor number.
- 34) ~~`fcreat()`~~ `fcntl()`
Performs various operations on file descriptors, such as duplication & locking.
- 35) `fsync()`
Writes all modified file data & metadata to permanent storage.
- 36) `fdatasync()`
Writes modified file data to permanent storage but may skip unnecessary metadata.
- 37) `ftruncate()`
Changes the size of a file using its path name.
- 38) `fttruncate()`
Changes the size of a file using its ~~path~~ name, file description.